A HYBRID BI-LSTM AND ENSEMBLE FRAMEWORK FOR ENHANCED HATE SPEECH DETECTION

Nawroz Haseen Tumul

Brac University

ABSTRACT

There has been a rapid and malicious spread of hate speech across all social media platforms in recent history, which have necessitated for the use of automated detection systems for content moderating as well as filtering as the sheer volume of content generated over social media makes it impossible for manual flagging of content. A significant challenge in this domain is the distinction of hate speech, which is malicious content targeted towards individuals or groups; and merely offensive speech. Even though existing research has made significant progress, many approaches still suffer from this exact same issue, which creates a lot of false positives, leading to content and users being unfairly filtered. In this paper, a hybrid machine learning architecture has been proposed to enhance the accuracy of this ternary classification task. The approach leverages an advanced feature extraction pipeline, which includes contextual embeddings from BERTweet, traditional TF-IDF & n-gram vectors as well well as linguistic features crafted from preprocessed texts. These features are processed by a two-stage pipeline that takes contextual learning from Bi-Directional LSTM and leverages that with ensemble classifiers like XGBoost and Random Forest for enhanced prediction. Evaluated on the widely-used Crowdflower Twitter dataset [1], this research's proposed model Hybrid Bi-LSTM - XGBoost achieves a state-of-the-art weighted F1-score of 95.50%, significantly improving the precision of identifying 'Hateful' speech compared to baseline models. Ablation studies have been used to verify the significant contribution of BERT embeddings and the extracted linguistic features. Furthermore, SHAP Explainable AI (XAI) has been used to explain the in-depth contribution of features for interpretability of the features in the hybrid models. The result demonstrated that both the Hybrid Bi-LSTM – XGBoost and Bi-LSTM – Random Forest architectures offer a superior performance and more interpretable solution for the classification of hate speech.

Keywords

Hate Speech Detection, Classification, Supervised Learning, Natural Language Processing, NLP, Sentiment detection, Hybrid Models, BERT, Deep Learning, Bi-LSTM, XGBoost, Random Forest. Explainable AI, XAI, SHAP

1. BACKGROUND AND MOTIVATION

1.1 Broad Context

There has been a rapid and malicious spread of hate speech across all social media platforms in recent history and with that the needs of content moderation through automated systems have become a necessity by means of algorithms and machine learning methods.[2] Especially considering the overarching reach of social media in the fields of news, global communication and politics, hate speech targeted towards specific race, gender, ethnicity, economic background can have real-world implications such as what we have witnessed in during the Rohingya Crisis, U.S. Capitol riots [4], which indicate a crucial need for such systems to prevent such social unrests. Secondly, regulatory directives such as the European Union's mandate that requires hate speech to be removed by social media platforms within 24 hours further reinforces this issue. [7]

1.2 The Problem

Automated hate speech detection, in spite of showing promise, is still an ongoing problem that remains a challenge do tackle, due to several factors including:

- Detecting hate speech almost always on relies on context, especially for things like sarcasm, slangs, banter and cultural references; which makes it difficult to distinguish between language that are hateful and language that is merely offensive or humorous. [3] [4]
- The issue of coded languages (e.g. 'b1tch', 'ret@rd' etc.) and misspellings (e.g. 'bich', 'kil' etc.) are tougher to detect.
- Context dependent slangs that are constantly evolving in social media. (e.g. 'woke', 'cooked', 'yolo' etc.)
- Annotated, licensed datasets that are available are often small, imbalanced and plagued by annotator bias, leading to state-of-the art models to perform well on specific datasets but suffer from performance drop when tested on cross-dataset benchmarks, highlighting this issue. [4]

1.3 Limitations of Existing Solutions

The existing approaches to this problem present a few limitations including:

- Keyword-based dictionaries and models like SVM or Naïve Bayes struggle with classifying hate speech as they fail to learn the proper contextual understanding of the data. [3] [4]
- While deep learning models like BERT and Bi-LSTM have significantly better performance compared to traditional machine learning models, they often rely on large, high-quality datasets but still face difficulties while dealing with code-mixed data. [2] [3]

1.4 Proposed Solution

To address the gaps, this research proposes a hybrid framework with in-depth preprocessing pipeline to extract diverse set of features such as semantic shifts, readability scores etc. to better learn deep syntactic and semantic understanding while also considering slangs, coded language and cultural biases. The framework also combines contextual embeddings from BERTweet as a feature to capture semantic nuances [1][4] on top of traditional TF-IDF and n-gram features. The hybrid architecture uses Bi-LSTM for sequence modeling, generating additional context on top of the existing features and combines them with ensemble classifiers like XGBoost and Random Forest to enhance the robustness and performance metrics of the models.

2. AIM AND RESEARCH QUESTION

The aim of this research is to address the issue of conflating hate speech with offensive language. The primary intuition for this limitation is the nuances of hate speech like context-dependent slangs, slurs that are used casually in a non-hostile way as well as sarcasm and coded language might me too complex to be captured by simplistic features alone (e.g. tweet length, keywords, basic n-grams etc.) alone. [1][5] To address this issue, research have shown that by incorporating more complicated features extracted from complex linguistic patterns that can capture nuances of language to better understand the syntactic and semantic understanding for the models.[6] Additionally, ensemble learning techniques such as XGBoost & Random Forest have shown promise in utilizing these complex features better by combining the learning from many decision trees in an efficient way, and have shown significant boost in performance when combined with deep learning models.[7][8] Our objective, therefore, is to develop an automated system that's more capable of capturing nuanced language for this ternary ('Hateful', 'Offensive', 'Neither') classification by utilizing a complex yet comprehensive feature set and combining them alongside hybrid architecture with ensemble techniques.

2.1 Key Objectives

- To implement a feature extraction pipeline to extract contextual embeddings from BERTweet, NLP features traditionally used for sentiment detection tasks like n-gams and TF-IDF vectors and well as linguistic features extracted from text structures and patterns like sentiment shifts, readability scores, NER, slang markers, lexical density etc. and metadata features like hashtag counts, mention counts to capture all the intricacies of the casual nature of tweets in general so that the ensemble techniques learn better. [7]
- To train and test various models on the dataset like classical ML models like Naïve Bayes, Logistic Regression, tree-based Ensemble models like Random Forest, XGBoost and Deep Learning methods like Bi-LSTM.
- To design hybrid models to combine the contextual learning of deep neural networks like Bi-LSTM with ensemble classifiers like Random Forest, XGBoost to test out how they compare in terms of performance and efficiency.
- In order to ensure that the models are both economical and effective. Complex preprocessing techniques like feature selection, normalization and hyperparameter tuning such as learning rates and tree depths are used to improve performance and lower cost of computation for the models.

- To conduct ablation studies, feature importance, per-class F1-scores and error analysis for each model to ensure which are the key contributing factors in terms of performance. Furthermore, to provide Global Model explainabilty using an Explainable AI like SHAP analysis to interpret the most influential feature to address the nature of complex models, which is critical to maintain AI ethics.[8]

2.2 This work is motivated by the following research question

Can a hybrid model architecture, by leveraging a rich suite of features, including contextual embeddings from BERT and deep linguistic patterns from in-depth preprocessing techniques, significantly improve the F1 score for identifying 'Hate Speech'; while effectively distinguishing it from 'Offensive Language' on a standardized, benchmark dataset?

2.2.1 Analyzing the Research Question

- The research question evaluates the efficacy of hybrid models (Bi-LSTM + XGBoost, Bi-LSTM + RF) against traditional and deep learning models, measuring by their f1 score to account for class imbalance, a common challenge in hate speech datasets [1]
- It focuses on specific gaps in performance in distinguishing between the 'Hateful' and 'Offensive' categories, which is one of the core challenges identified.[1]
- It assesses the value of a comprehensive feature sets, moving past the basic patterns of tweet lengths, word counts and unigrams used in earlier works to include deep contextual representations in multiword expressions.[6]
- The requirement for explainability (such as explainable AI like SHAP) ensures that the models' decisions are interpretable, addressing a concern for ethical deployment of such pattern recognition systems to help mitigate potential biases.[8][9]

2.3 The Value of the Research

The value of this research is twofold. Academically, it contributes a hybrid machine learning framework that can be used to provide empirical evidence on the impact of different, fine-tuned feature sets for a natural language processing task of sentiment classification- that can be used as benchmark for future studies. Practically, on the other hand, this research proposed a methodology that can be used for potential improvement for content moderation in online platforms to reduce potential threats of harmful content while still being able to minimize the unjust false alarms for users who might just be using offensive language for communication- thereby enhancing the environment of online discourse.

3. METHODOLOGY

This research is a classification task that falls under the domain of natural language processing (NLP) that aims to take raw text media and to categorize those into 'Hateful', 'Offensive' or 'Neither'. A fundamental task for model training in NLP is a clear and reproducible preprocessing pipeline, which is critical for transforming raw, noisy social texts into features that are semantically and syntactically meaningful. The process involves several steps to ensure the quality and relevance of the data is maintained within the context that they were used- a practice emphasized in NLP research to mitigate noise and bias that are common in user generated content. [9]

The research pipeline followed was Data Collection (Crowdflower Dataset [1]), Exploratory Data Analysis, Data Preprocessing, Feature extraction, Data Splitting (80% Training, 20% Test data), Model Selection & model training. Based on the results of the Models on the training set, models were chosen for

the Hybrid architecture, then the hybrid architectures were trained.Finally, all the models were tested and compared.

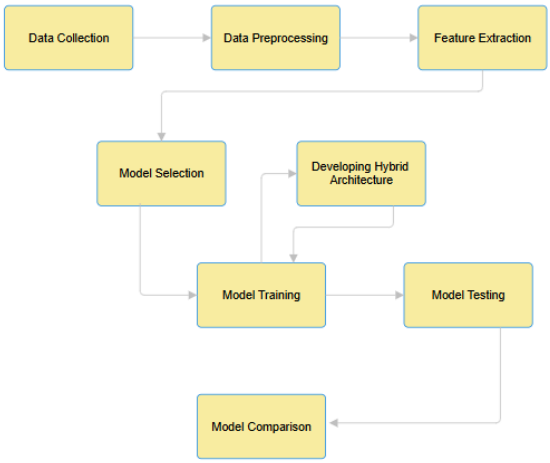


Figure 1 Pipeline for this research

3.1 Exploratory Data Analysis

An exploratory analysis of the Crowdflower dataset (Davidson et al., 2017)[1] confirmed its integrity, revealing no missing values across all 24,783 tweets.

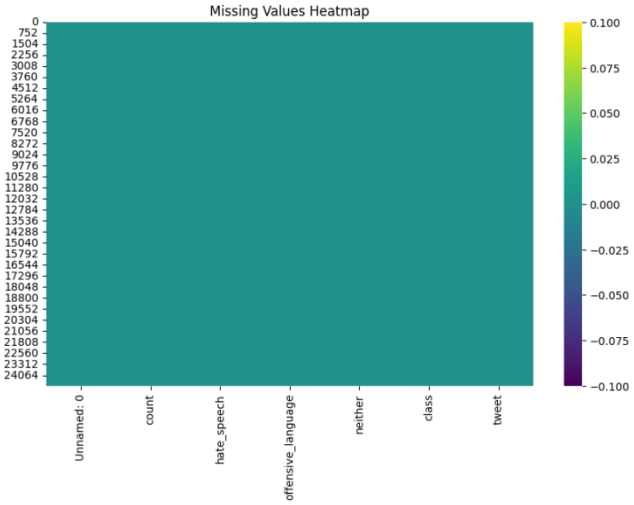


Figure 2 Missing Value Analysis

A significant class imbalance was identified, with "Offensive Language" (77.4%) dominating the dataset, while "Hate Speech" constituted a small minority (5.8%), necessitating the use of F1-score for robust evaluation.

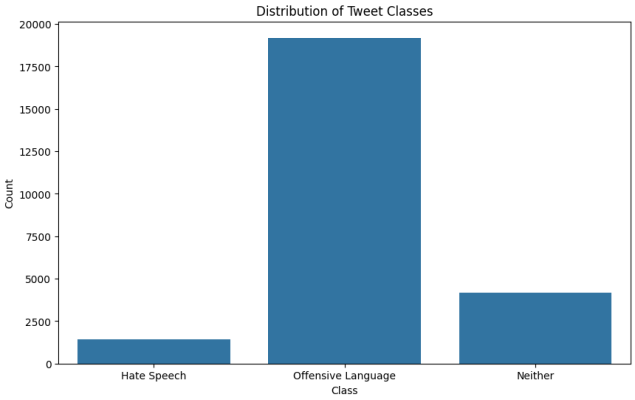


Figure 3.1 Class Distribution Bar Chart

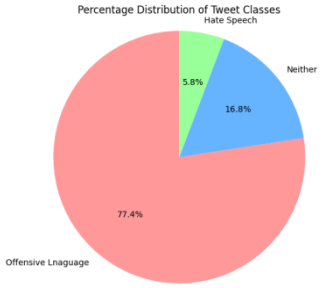


Figure 3.2 Class Distribution: Pie Chart

Linguistic feature analysis showed "Hate Speech" tweets used significantly fewer hashtags but more mentions than other classes, suggesting a more targeted nature.

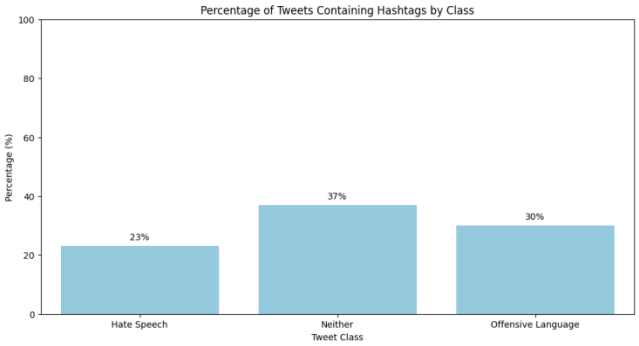


Figure 4.1 Hashtags by class

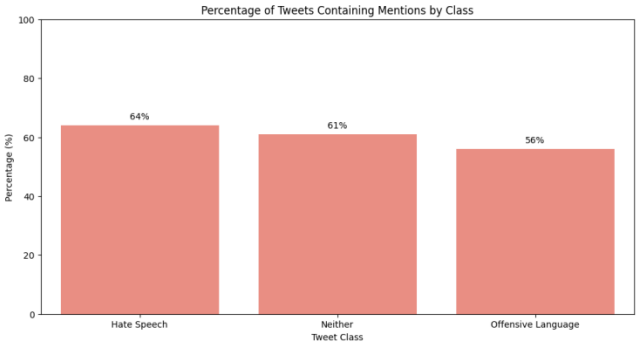


Figure 4.2 Mentions by class

Tweet length analysis found a minimal number of outliers (<0.3%), primarily in the "Offensive" class.

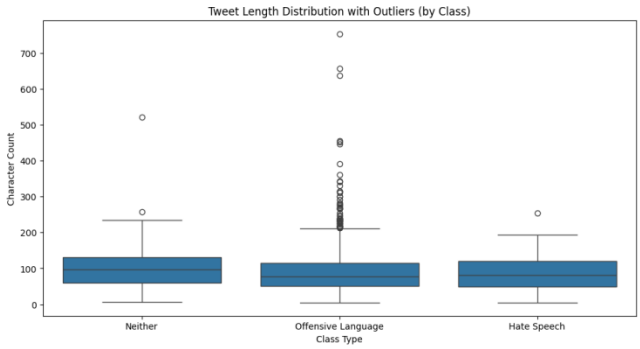


Figure 5.1 Tweet length distribution outliers: boxplot

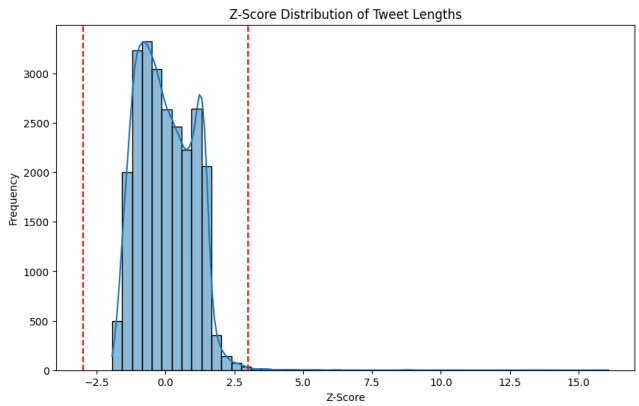


Figure 5.2 Tweet length distribution: Z-Score

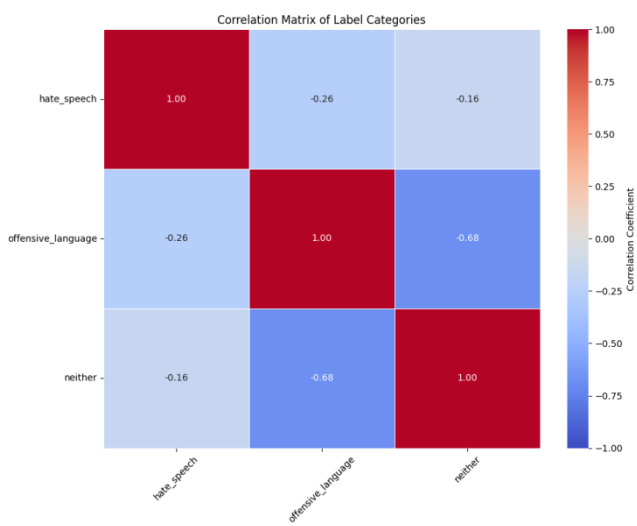


Figure 6 Correlation Matrix of class labels

Aspects Analyzed	Key Finding	Implication for Modeling
Missing Data	No missing values in any column.	No data imputation required; full dataset is usable.
Class Distribution	Highly imbalanced: 77% Offensive, 17% Neither, 6% Hate.	Mandates use of F1-score, precision, recall for evaluation. Potential need for stratified sampling or class weighting. Confirms the nuanced, subjective nature of the task, aligning with established research.
Annotation Subjectivity	Sum of individual labels > number of tweets.	Confirms the nuanced, subjective nature of the task, aligning with established research.
Linguistic Features	Hate Speech tweets use fewer hashtags but more mentions.	Mentions are a potentially important feature for identifying targeted, hateful content.
Tweet Length	Minimal outliers (0.24%), mostly in "Offensive" class.	Outliers are quite few in terms of the scale of the dataset.

Table 1: Key Findings in EDA

This analysis confirms the dataset's quality while highlighting its primary challenge (imbalance) and opportunities (linguistic feature engineering) for the subsequent modeling phase.

3.2 Data Pre-processing

This research followed a preprocessing pipeline to transform the raw, noisy tweets into clean, structured format suitable for feature extraction and model training, the pipeline consisted of several methodical steps to ensure semantic and syntactic quality of the data.

Text Cleaning: Raw tweets were cleaned by removing user mentions (@'username') and hashtags (#'topics'), URLs and non-alphabetical characters using regular expressions, this step was crucial for keeping the core linguistic content of the tweets intact.

Outlier Handling: Tweet length outliers, as previously identified in the EDA using Z-scores (threshold: |3.5|), were removed as this minimally affected the size of the dataset to ensure the performance of models were stable.

Addressing Class Imbalance: To address the severe class imbalance as observed from the EDA, a combination of undersampling the majority class 'Offensive Language' and oversampling of the minority class "Hate Speech" was applied to make sure a more balanced training set of 16,017 samples, with a distribution of 9,000 (Offensive), 4,159 (Neither), and 2,858 (Hate Speech).

Text Normalization: The cleaned text was tokenized into individual words, which were then lemmatized- which is basically used to reduce words to their base form in the dictionary (e.g. 'running' to 'run') using the WordNet Lemmatizer, in combination with parts-of-speech tagging for accuracy. Finally, stopwords (e.g. 'the', 'and', 'is' etc) were removed to eliminate tokens that appear in most tweets as they don't contain any meaningful information.

The output of this preprocessing pipeline was a corpus of clean, lemmatized tokens.

3.3 Feature Extraction

3.3.1 Features extracted

The set of features extracted focused on the task of capturing as much of the lexical, semantic, syntactic and stylistic learning from the tweets that are by nature very casual and don't follow strict grammatical rules, which traditional natural language processing features like bag-of-words won't be able to sufficiently capture.[2]

Lexical & Semantic Features:

1. TF (Term Frequency for each token all the tweets) and IDF (Inverse Document Frequency; the amount of time a token has appeared across all individual tweets) has been encoded as a matrix consisting of vectors (max_features = 5000) to determine the importance of each of the tokens- tokens which appeared across all tweets will not be as important.
2. N-gram matrix (max_features = 2000), exclusively bigrams and trigrams were used to capture expressions & phrases.[12]
3. The pretrained (on Twitter data) transformer model BERTweet was used to generate 768-dimensional embeddings per tweet in order to encode contextual relationships between words.[13]

Psycholinguistic features like the following were also extracted as well, as a method to capture structural and syntactic knowledge of the texts as well as the semantic understanding of the writing style and aggression markers.[5]

- Structural Features:
  4. Cleaned Tweet Length: Total character count of preprocessed, clean tweets. Tweets that are longer or shorter can be indicative of different sentiments; longer tweets may be an indication of arguments and dispute.
  5. Average Word Length: Mean of the characters per word in tweets. Shorter words might indicate informal language and slangs.
- Metadata Features:
  6. Hashtag Count: Fewer hashtags might be indicative of regular tweets, while multiple hashtags could indicate grouped altercations.
  7. Mention Count: A higher mention count could indicate targeted harassment directed at specific individuals
- Sentiment Features:
  8. VADER Sentiment Score: A compound score quantifying tweet polarity (negative to positive) [10].
  9. Lexical Density: estimates lexical word ratio in tweets. Lower density may indicate repetitive and slang filled language.
- Readability Features:
  10. Flesch-Kincaid: estimates the grade level required to understand a text, useful for hate tweets as those often use simpler, more direct language
  11. Flesch-Ease: Scores text based on ease-of-reading. Lower scores indicate more complex text; which could be used to determine hate masked in sophisticated rhetoric
  12. Gunning Fog: Determines the years of formal education needed to interpret text. Also used to distinguish direct insults and veiled hate.
- Slang and Dialect Features
  13. Slang Marker: The frequency of words matching custom regular expression patterns for misspellings (e.g. 'botch', 'retsrd' etc.) and coded slurs (e.g. 'id10t', '@$$' etc.). A high count is a strong indicator of offensive and hateful language.
- Named Entity Recognition Features
  14. Race marker: frequency of words related to race, nationality or ethnicity
  15. Religion Marker: frequency of words related to religious groups
  16. Figure Marker: frequency of mentions of public figures

  These could be used to find hateful tweets directed at individuals or groups [1].
- Parts of Speech Features:
  17. Aggressive verb-noun combinations: count of aggressive verbs (e.g. 'kill', 'hate') followed by a noun or pronoun (e.g. 'kill him', 'hate police' etc.). This pattern often indicated direct threat in tweets.[9]

3.3.2 Feature Scaling

The numerical features extracted from texts (features 4-17) were scaled using a Standard Scaler and made into a feature group named 'Numerical Features' for better interpretability of the model training.[11] This other feature groups used were 'BERT embeddings', 'TF-IDF Matrix' and 'n-gram Matrix', which, combined with 'Numerical Features' group were combined into a sparce matrix called 'combined_features' with 7,787 dimensions per sample.

3.3.3 Dealing with Class Imbalance

After features extraction, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the high-dimensional feature space to balance the class distribution, generating 9,000 samples per class for robust model training.[14]

3.4 Individual Model Approach

A select few models have been used to test out how well they train on the dataset and their performance (F1-scores) on each individual features. The research denotes the performance of probabilistic models like Naïve Bayes, linear discriminative classifier like Logistic Regression, ensemble classifiers like Random Forest & XGBoost as well as Deep Learning models like Bi-LSTM for this evaluation process before moving on to designing and testing out hybrid architectures.

3.4.1 Using Naïve Bayes

Naïve Bayes is a probabilistic classifier, that's very popular in ML studies because of its extremely efficient training time on even medium to large datasets while still giving decent performance metrics. The way that it's able to achieve such efficiency is because of the assumption of feature independence- which simplifies the data significantly while increasing efficiency.[11] For this task, the research moved forward with this model with the intuition that it will be able to train on the very large and sparse matrices of the features like TF-IDF, n-grams; both having 5000 and 2000 dimensions respectively. This makes it suitable for initial assessment of the dataset's quality for establishing a baseline for performance going forward with the other models.

3.4.2 Using Logistic Regression

Logistic regression is used to test out the feature set because of its high interpretability to quantify the relationships between the extracted features (e.g. slang markers, sentiment shifts etc.) and their impact on each of the classes. Using L1/L2 regularization also helps preventing overfitting issue on the feature set, which makes it a reliable method for classifying natural language processing tasks like this.[9] Futhermore, Logistic Regression is trained using Maximum Likelihood Estimation, which is often optimized using iterative algorithms like L-BGS to make sure it converges, even with high dimensional data.

3.4.3 Using Random Forest

This ensemble method was utilized for its superior ability to model complex, non-linear decision boundaries by constructing a multitude of de-correlated decision trees. The model uses a multitude of decision trees and uses the mode of their predictions as output of classification tasks like these. Each of the trees use something known as 'Bootstrap Sampling', the method of how each of these trees train on a random subset of the data, and in each split a random subset of features is considered to make sure the trees are de-correlated. Then, the predictions from these are then aggregated the mode is calculated for classification tasks. [14] Random Forest is robust to outliers and noise, as well as having the capability to capture interactions between all the features and their impact on the performance making it one of the primary ensemble techniques for this research.

3.4.4 Using XGBoost

XGBoost is used in this research as another ensemble framework to evaluate performance on the feature set. This model works by using decision trees similar to Random Forest, but the difference is in XGBoost, the decision Trees learn from the residual errors, the difference between current predictions and the actual values of the previous trees and builds upon it by minimizing a defined loss function (like log loss for classification tasks) using gradient descent, making it useful for structural tasks similar to this research. Furthermore, by incorporating L1 and L2 regularization, similar to Logistic Regression, the model also prevents overfitting. Additionally, by using techniques like tree pruning and cross-validation, XGBoost can handle missing values (wasn't necessary for this research dataset) and also support parallel processing. These, on top of the fact that the model works well with structural data and its high interpretability made XGBoost another valid ensemble model for this research.

3.4.5 Using Bi-Directional LSTM (Long Short-Term Memory)

This is a recurrent neural network architecture that works extremely well for sequential text data to capture the contextual nature of the language and it was selected because of it. Unlike traditional NLP approaches like bag-of-words models, the Bi-LSTM works by processing the text in both forward and backward direction, thus better capturing the long-range dependencies and full context of the words used.[16][17] Unlike the models that use the bag-of-word approach, which treats words as unordered sets of tokens[18], Bi-LSTM maintains sequential integrity, this is why it is a particularly powerful option in understanding sarcasm, negation and other complex syntactic structures that are crucial for distinguishing between 'Hateful' and 'Offensive Language', which was a key driving factor for this research. The model employs gated mechanisms such as input, forget and output gates to regulate the flow of information, all the while minimizing the vanishing gradient problem, enabling learning sequences that are over extended. [17]

3.5 Hybrid Model Approach

The research proposes a hybrid ensemble framework to synergistically combine the strength of deep learning and traditional machine learning approaches. The intuition came from the observation that even though Bi-LSTMs excel at capturing complex sequential patterns and textual contexts, tree-based models like XGBoost and Random Forest demonstrate better performance in learning hierarchical decision boundaries from structured feature representation. The aim of this hybrid methodology is to leverage the complementary strengths of these architectures for enhanced hate speech detection.

3.5.1 Hybrid Bi-LSTM + XGBoost/Random Forest

The hybrid approach operates through a two-stage pipeline. First, a Bi-LSTM feature extractor processes the input sequences through two bidirectional LSTM layers to capture temporal dependencies and contextual representations, followed by two dense layers to find complex patterns. The model architecture specifically extracts 64 features, learned from the penultimate layer (before the final classification dense layer), preserving high-level semantic patterns while reducing dimensionality. After learning these features, they are concatenated with the original, combined features (TF-IDF matrix (5000), n-gram matrix (2000), BERT embeddings (768), numerical features (19); combined 7787 features) for creating a rich feature space that captures both sequential and structural information.

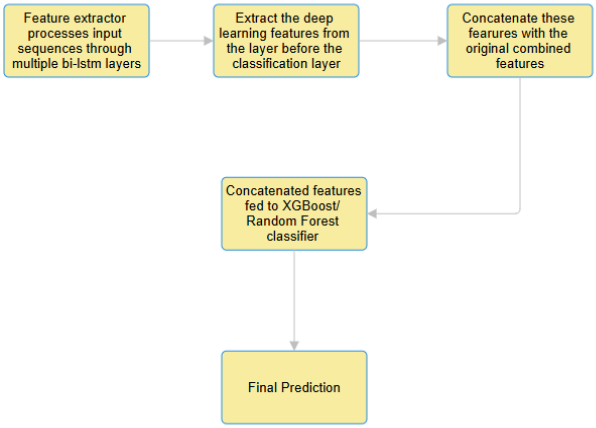


Figure 7 Hybrid Bi-LSTM + XGBoost/Random Forest Pipeline

After that, the concatenated features are fed into XGBoost or Random Forest classifiers for the final prediction. The XGBoost implementation utilizes histogram-based tree methods that correct errors sequentially for computational efficiency, while the Random Forest classifier incorporates class weighting to address residual imbalance in the independent trees by averaging their predictions.

3.5.2 Methodological advantages

This hybrid model addresses several limitations of the standalone models:

1. It compensates for the Bi-LSTM's lack of interpretability due to its 'Black Box' nature by introducing highly interpretable models for the final classification

2. It combines the contextual embeddings from deep-learning with the ensemble models' learning of the extracted features- this way, the hybrid architecture takes the best of both models.

By using this approach, contextual distinctions can be made between hate speech and offensive language more effectively as Bi-LSTM models capture sequential data very well, which includes negation patterns (e.g. "This movie is not good and is actually quite boring"- traditional features like n-grams would look at the token 'good' and assign the text some positive value because of it. But because Bi-LSTM does two passes for each sentence, forward and backward, it will capture 'not' before 'good' in both passes, making it accurately score the sentence as negative and not somewhat positive). With addition to the ensemble models working on the extracted features, for example, sentiment shift, which assesses a tweet's shift in sentiment- this combination would provide for a better understanding of the data.

3.5.3 Implementation Details

As previously mentioned, the Bi-LSTM model used 4 layers: 2 Bi-LSTM layers (128 and 64 units respectively) followed by 2 Dense layers (53 and 32 units respectively) using ReLU activation and dropout (0.2-0.3) for training on the raw features, which were standardized and reshaped to [samples, 1, features] for compatibility. The training had a maximum of 50 epochs using Adam optimizer with a Learning Rate of 0.001 and the Loss Function used was Cross-entropy Loss, with Early Stopping (patience-8) and Learning Rate Reduction (factor=0.2, patience=5). The contextual features were extracted from the output of the 32-unit Dense layer (model.layers[-3]); after which these 64-dimensional embeddings were concatenated with the original combined scaled 7787-dimensional features to create a scaled 7851-dimensional hybrid feature set for the tree-based classifiers. Finally, for the an XGBoost classifier (n_estimators=100, max_depth=6) and a Random Forest classifier (n_estimators=100, class_weight='balanced') were optimized on this hybrid feature set for the final prediction.

4. MODEL EVALUATION

For model evaluation, performance metrics such as accuracy, precision, recall and F1-scores, confusion matrices for the representation of classification performance, ablation study and feature importance for importance of specific feature groups and class-wise F1 scores and error rates were calculated.

4.1 Individual Models

4.1.1 Naïve Bayes

Naïve Bayes established a strong baseline with a test accuracy of 83.59%; and precision, recall, F1-score of 85.20, 83.59% and 83.22% respectively. It also shows a high AUC score of 87.83%. It showed excellent recall for hate speech (98%) but lower precision (75%), which indicates some false positives in this category.

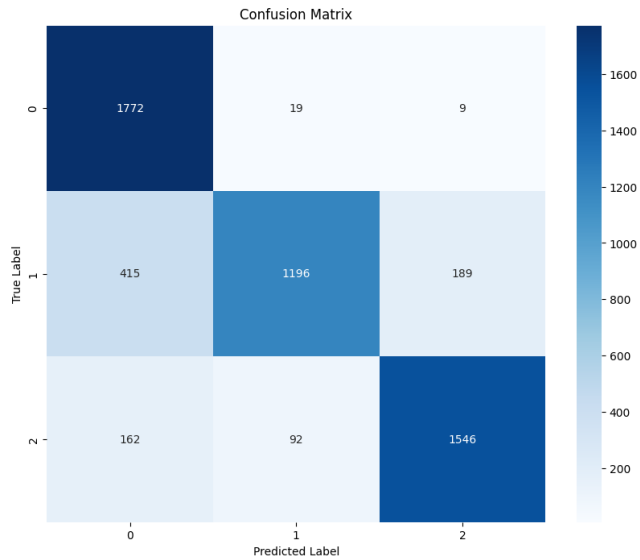


Figure 8.1 Confusion Matrix of Naïve Bayes

From the ablation study, it shows that BERT embeddings were the most impactful feature group for Naïve Bayes, achieving an F1-score of 0.6914, significantly outperforming TF-IDF (0.6363), N-grams (0.5963), and numerical features (0.5376).

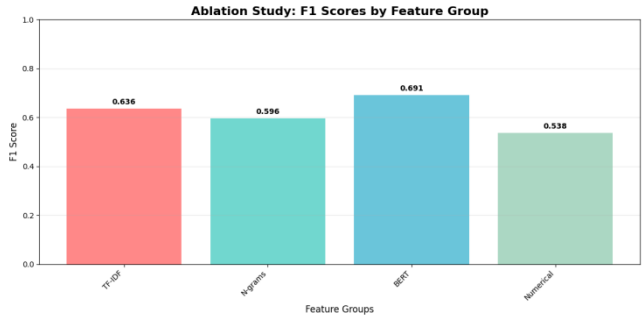


Figure 8.2 Ablation Study of Naïve Bayes

Naïve Bayes performed well on Class 0 (Hateful: F1=0.8542) and Class 2 (Neither: F1=0.8725), while struggling more with Class 1 (Offensive: F1=0.7699), as indicated by the class wise f1 score.

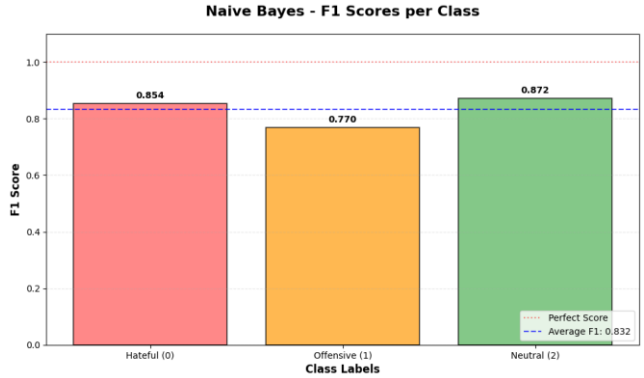


Figure 8.3 F1 scores per class of Naïve Bayes

Error analysis revealed the lowest error rate for Class 0 (1.56%), with higher error rates for Class 1 (33.56%) and Class 2 (14.11%), indicating particular difficulty in distinguishing offensive language from other categories.

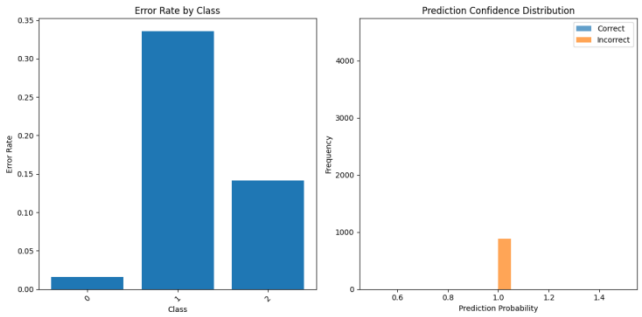


Figure 8.4 Error Rate by class of Naïve Bayes

4.1.2 Logistic Regression

Logistic Regression demonstrated superior performance with a test accuracy of 93.52%, as well as precision, recall, f1 and AUC scores of 93.64, 93.52, 93.44 and 97.64 respectively, significantly outperforming the Naïve Bayes baseline. Additionally, it shows better precision (91%) for hate-speech as well.

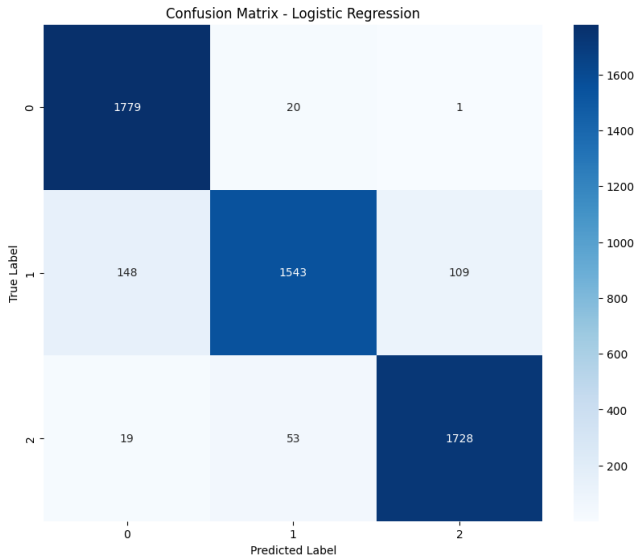


Figure 9.1 Confusion Matrix of Logistic Regression

The ablation study revealed that TF-IDF features were the most impactful for this model, achieving an exceptional F1-score of 0.9878, substantially outperforming BERT embeddings (0.8687), N-grams (0.7879), and numerical features (0.5879).

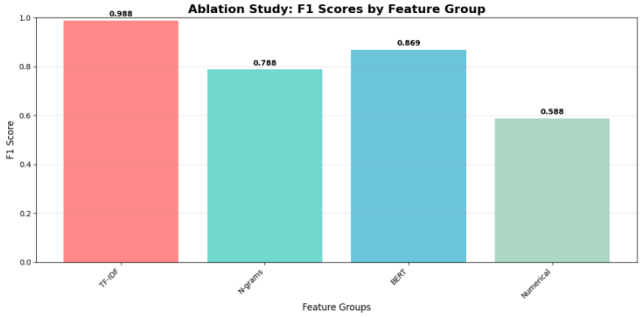


Figure 9.2 Ablation Study of Logistic Regression

Class-wise performance was notably balanced with strong F1-scores across all categories: Class 0 (Hateful: 0.9498), Class 1 (Offensive: 0.9034), and Class 2 (Neither: 0.9500).

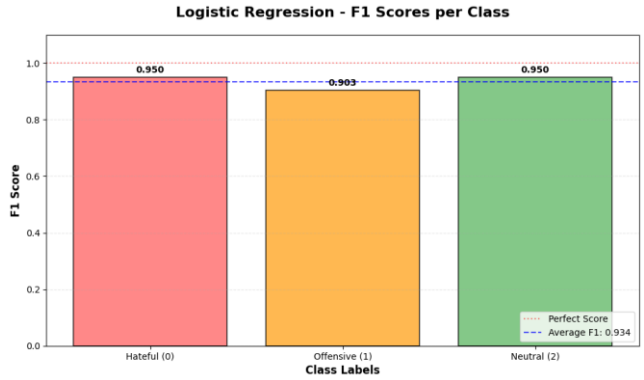


Figure 9.3 F1 Scores per Class of Logistic Regression

Error analysis showed significantly improved performance with the lowest error rates observed: Class 0 (1.17%), Class 1 (14.28%), and Class 2 (4.00%).

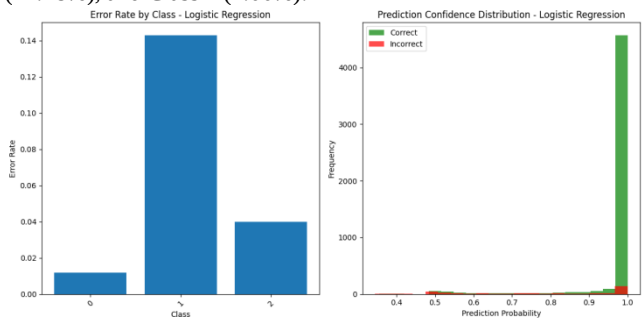
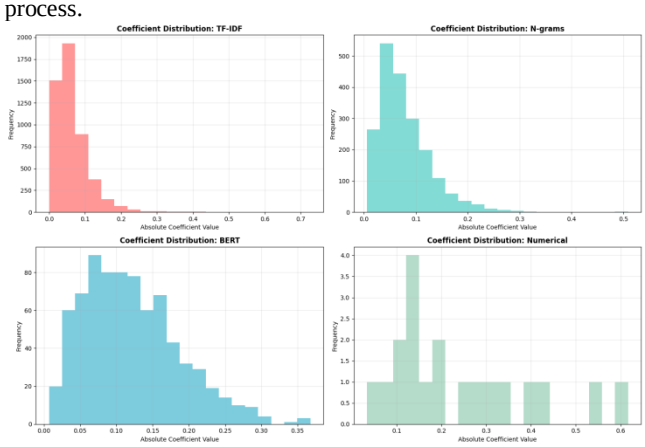


Figure 9.4 Error Rates per Class (left), Prediction Confidence Distribution of Logistic Regression (right)

Feature importance analysis revealed that numerical features carried the highest average absolute coefficients (0.2381), followed by BERT embeddings (0.1214), suggesting these feature groups were most influential in the model's decision-making process.



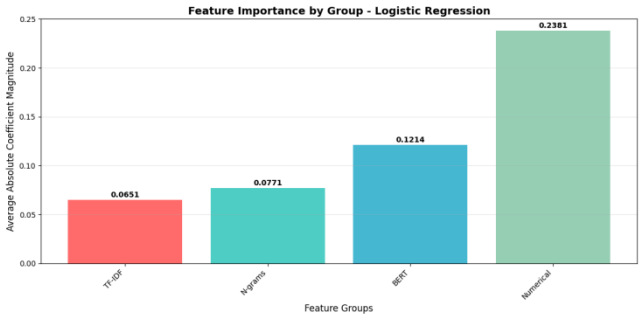


Figure 9.5 Coefficient Distribution per Feature Group (above) Feature Importance by Group of Logistic Regression (below)

4.1.3 Random Forest

Random Forest achieved exceptional performance with a test accuracy of 94.11%, representing the strongest results among the individual models evaluated. The model demonstrated outstanding overall metrics with precision of 0.9423, recall of 0.9411, F1-score of 0.9413, and a near-perfect AUC score of 0.9904. Also, better precision for hate-speech (99%).

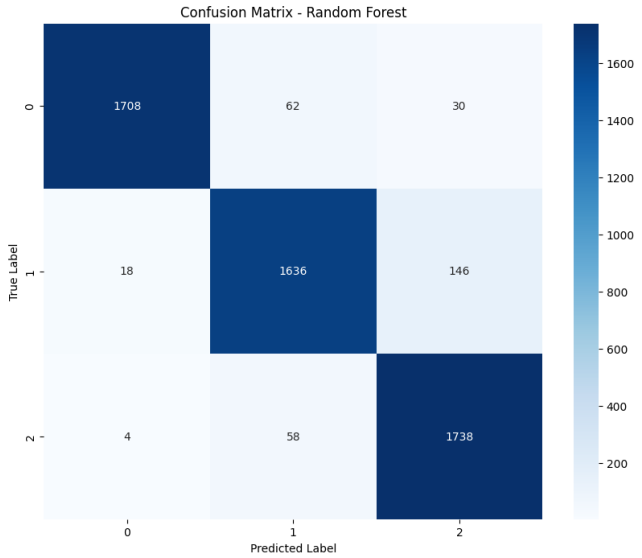


Figure 10.1 Confusion Matrix of Random Forest

The ablation study showed exceptional performance with multiple feature sets, with BERT embeddings achieving a remarkable F1-score of 0.9989, closely followed by numerical features (0.9987), TF-IDF (0.9952), and N-grams (0.8669).

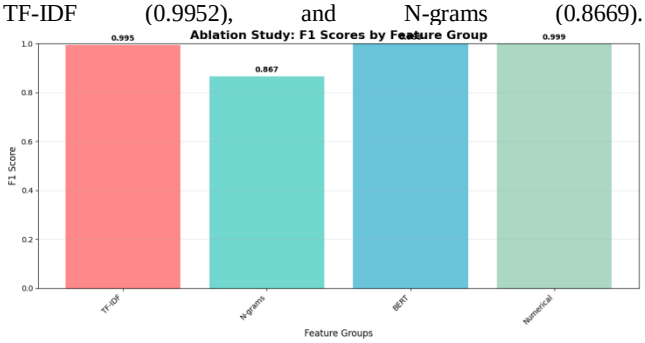


Figure 10.2 Ablation Study of Random Forest

Class-wise performance was particularly impressive for hate speech detection (Class 0: F1=0.9677), while maintaining strong results for offensive language (Class 1: F1=0.9201) and neutral content (Class 2: F1=0.9359).

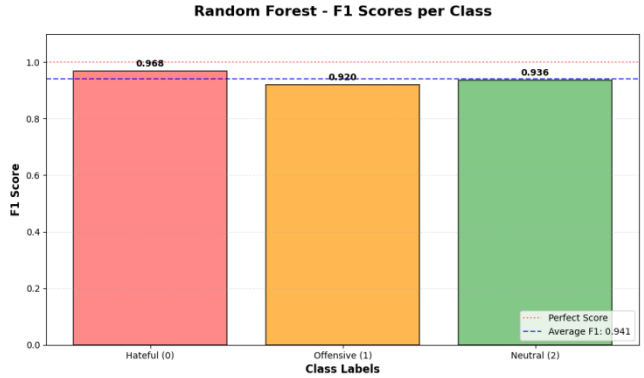


Figure 10.3 F1 Scores per Class of Random Forest

Error analysis showed the lowest overall error rates among all models, with Class 2 achieving exceptional performance (3.44% error rate), while maintaining strong results for Class 0 (5.11%) and Class 1 (9.11%).

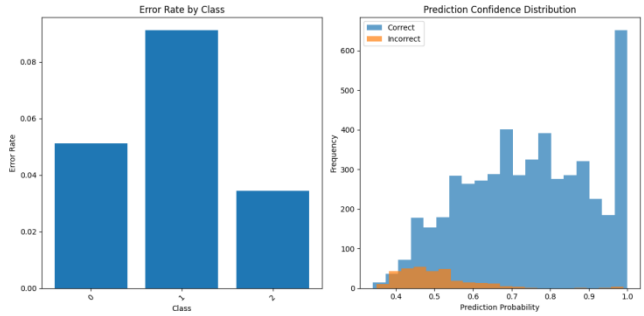


Figure 10.4 Error Rate by Class (left) and Prediction Confidence per class (right) of Random Forest

Despite the model's excellent performance, feature importance analysis revealed relatively low average importance scores across all feature groups, with numerical features (0.0014) and BERT embeddings (0.0012) showing minimal but comparable importance, while TF-IDF and N-grams registered negligible importance values, suggesting the model effectively distributed importance across many features rather than relying on dominant feature groups.

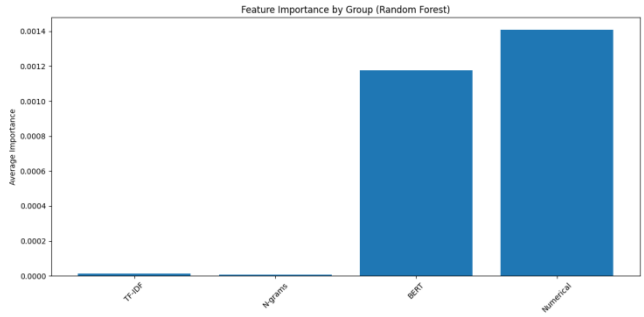


Figure 10.5 Feature Importance by Group of Random Forest

4.1.4 XGBoost

XGBoost delivered strong performance with a test accuracy of 93.44%, demonstrating results comparable to the top-performing

individual models. The model achieved excellent overall metrics with precision of 0.9354, recall of 0.9344, F1-score of 0.9340, and a high AUC score of 0.9878, slightly lesser precision (94%) than Random Forest for hate speech.

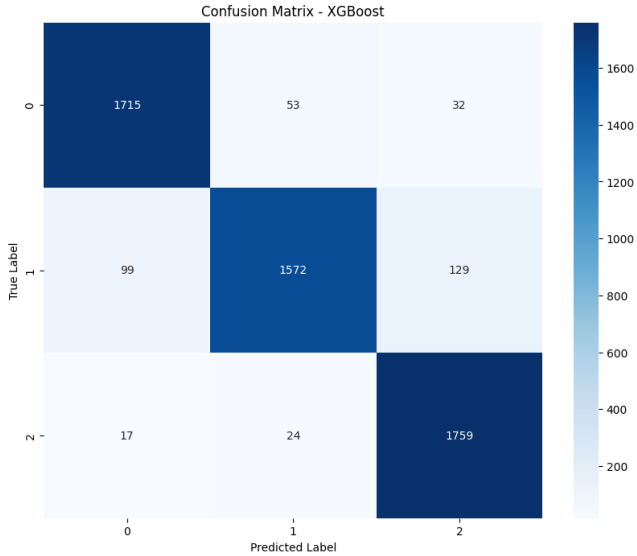


Figure 11.1 Confusion Matrix (XGBoost)

The ablation study revealed BERT embeddings as the most impactful feature set with an F1-score of 0.9269, significantly outperforming TF-IDF (0.8433), numerical features (0.6884), and N-grams (0.5545).

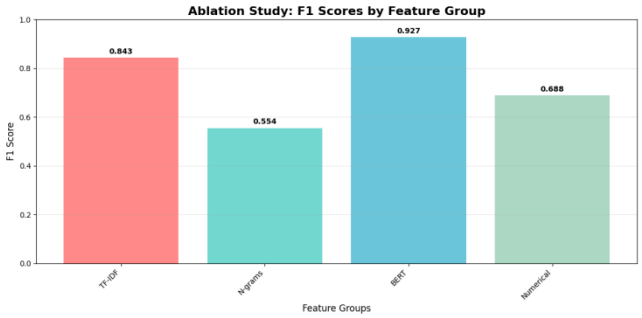


Figure 11.2 Ablation Study (XGBoost)

Class-wise performance showed strong results across all categories with Class 2 (Neither) achieving the highest F1-score (0.9457), followed by Class 0 (Hateful: 0.9446) and Class 1 (Offensive: 0.9116).

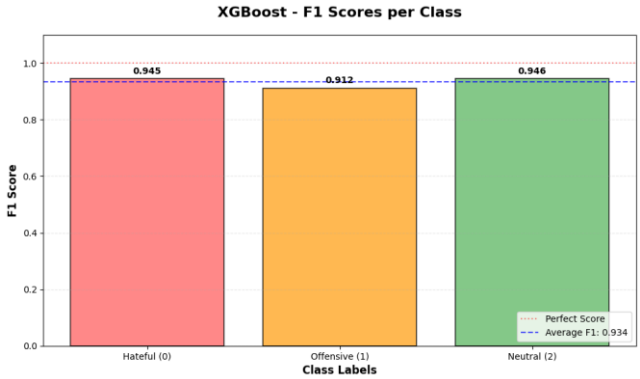


Figure 11.3 F1 Scores per Class (XGBoost)

Error analysis revealed impressive performance for Class 2 with only a 2.28% error rate, while Class 0 (4.72%) and Class 1 (12.67%) showed slightly higher but still strong error rates.

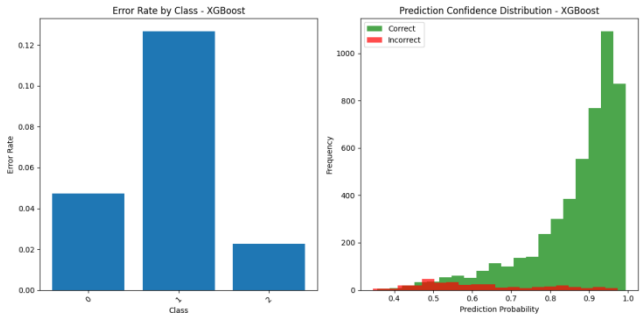
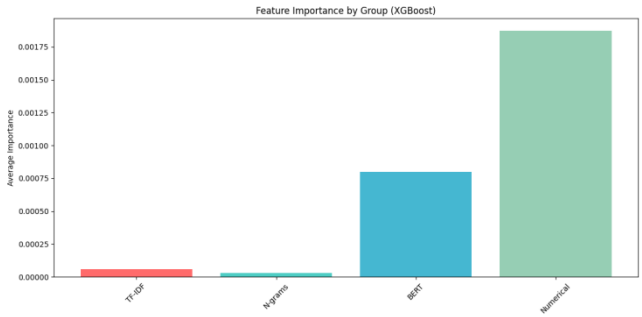


Figure 11.4 Error Rate by Class (left) and Prediction Confidence accuracy of XGBoost (right)

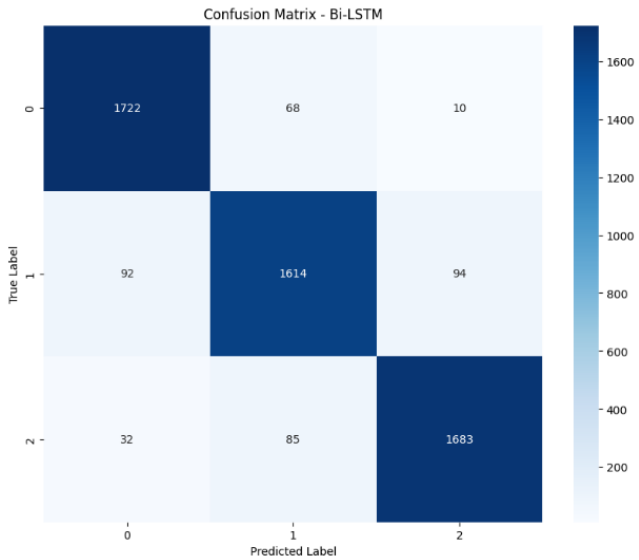
Feature importance analysis indicated that numerical features carried the highest average importance (0.0019), followed by BERT embeddings (0.0008), with TF-IDF and N-grams showing minimal importance, suggesting the model primarily leveraged the hand-crafted and contextual embedding features for its decisions.



11.5 Feature Importance by Group (XGBoost)

4.1.5 Bi-LSTM

The BiLSTM model demonstrated strong performance with a test accuracy of 92.94%, showcasing the effectiveness of deep learning for sequential text classification. The model achieved well-balanced metrics with precision of 0.9293, recall of 0.9294, F1-score of 0.9293, and a high AUC score of 0.9865. The model demonstrated particularly strong recall for hate speech (96%), indicating effective identification of true positive cases in this critical category, while maintaining solid precision (93%) for hate speech.



12.1 Confusion Matrix (Bi-LSTM)

Class-wise performance was particularly notable for hate speech detection (Class 0: F1=0.9446), while maintaining strong results for neutral content (Class 2: F1=0.9384) and offensive language (Class 1: F1=0.9050).

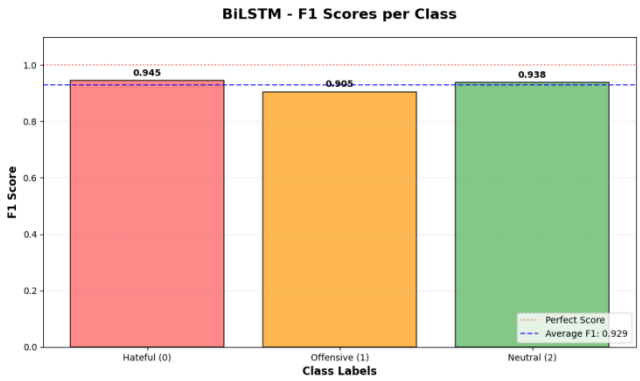


Figure 12.2 F1 Scores per class (Bi-LSTM)

Error analysis revealed excellent performance across all classes, with Class 0 showing the lowest error rate (4.33%), followed by Class 2 (6.50%) and Class 1 (10.33%).

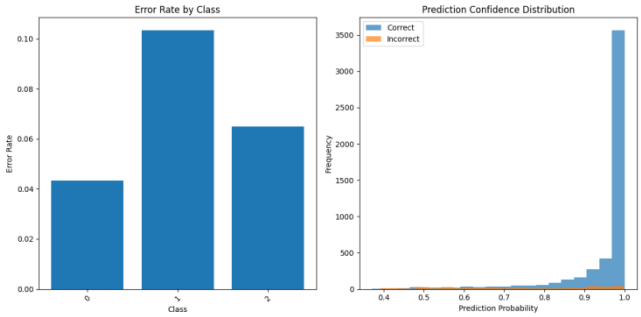


Figure 12.3 Error Rate by class (left), Prediction Confidence Distribution of Bi-LSTM (right)

4.2 Hybrid Models

4.2.1 Hybrid Bi-LSTM - XGBoost

The hybrid BiLSTM-XGBoost architecture demonstrated exceptional performance, achieving the highest test accuracy of 95.50% among all evaluated models, representing a significant improvement over individual model. The hybrid approach achieved outstanding overall metrics with precision of 0.9549, recall of 0.9550, F1-score of 0.9550, and a near-perfect AUC score of 0.9915. A state-of-the-art 97% precision, recall & F1-score for hate-speech is also a very positive outcome for this research as a whole.

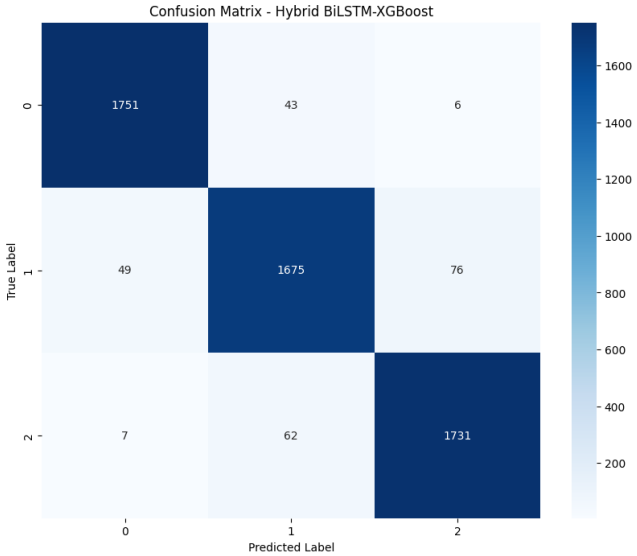


Figure 13.1 Confusion Matrix (Bi-LSTM – XGBoost)

Class-wise performance was remarkably balanced and superior across all categories, with Class 0 (Hateful) achieving an exceptional F1-score of 0.9709, followed by Class 2 (Neither: 0.9582) and Class 1 (Offensive: 0.9358).

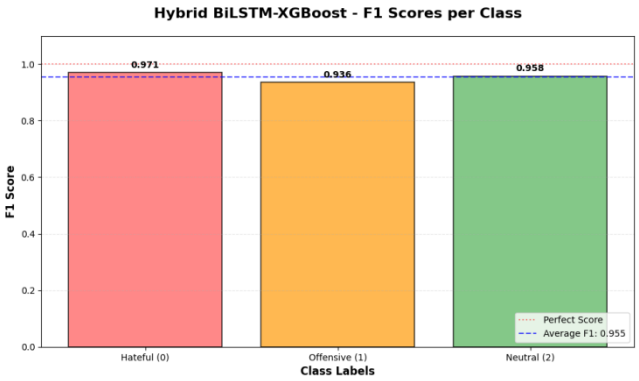


Figure 13.2 F1 Scores per Class (Bi-LSTM – XGBoost)

Error analysis revealed the lowest error rates observed in the study, with Class 0 showing minimal errors (2.72%), while Class 1 (6.94%) and Class 2 (3.83%) also demonstrated substantial improvements over individual models.

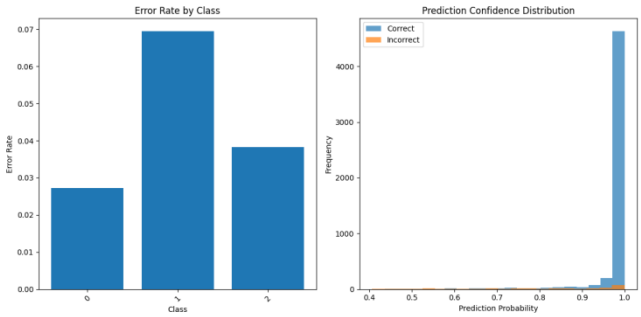


Figure 13.3 Error Rate by Class (left), Prediction Confidence Distribution of Bi-LSTM – XGBoost (right)

Feature importance analysis showed relatively balanced but minimal importance scores across feature groups, with both numerical features and BERT embeddings contributing equally (0.0003), suggesting the hybrid model effectively synthesized information from multiple feature types to achieve its superior performance.

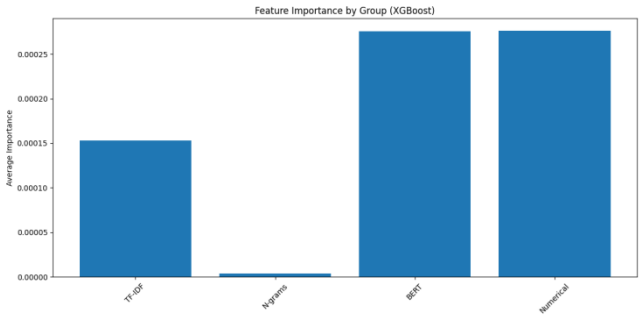


Figure 13.4 Feature Importance by Group (Hybrid XGBoost)

4.2.2 4.2.2 Hybrid Bi-LSTM - Random Forest

The hybrid BiLSTM - Random Forest architecture delivered exceptional performance, achieving a test accuracy of 95.17%, making it the second highest-performing model and demonstrating the effectiveness of combining deep learning with ensemble methods. The model achieved outstanding overall metrics with precision of 0.9516, recall of 0.9517, F1-score of 0.9516, and a remarkable AUC score of 0.9922 - the highest AUC observed in the study. It gave a very slightly less (96%) precision for hate speech but overall almost equal recall & F1-score as the hybrid XGBoost model.

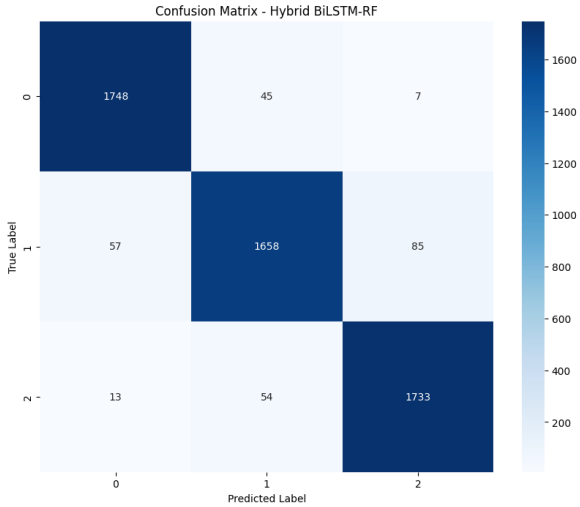


Figure 14.1 Confusion Matrix (Bi-LSTM – Random Forest)

Class-wise performance was excellent across all categories, with Class 0 (Hateful) achieving an F1-score of 0.9663, followed by Class 2 (Neither: 0.9561) and Class 1 (Offensive: 0.9322).

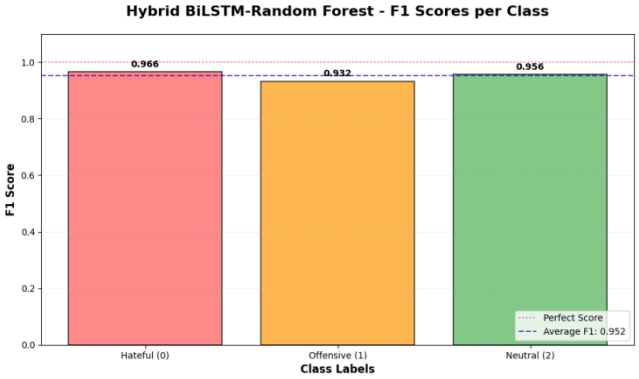


Figure 14.2 F1 Scores per class (Bi-LSTM – Random Forest)

Error analysis revealed superior performance with Class 0 showing minimal errors (2.89%), while Class 1 (7.89%) and Class 2 (3.72%) demonstrated significant improvements over individual models.

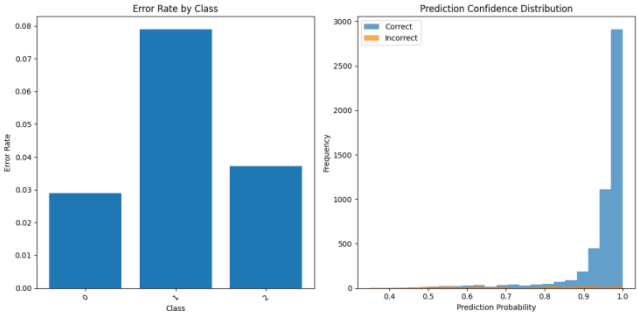


Figure 14.3 Error Rate by Class (left), Prediction Confidence Distribution of Bi-LSTM – Random Forest (right)

Feature importance analysis showed balanced contributions across feature groups, with numerical features (0.0003) slightly outperforming BERT embeddings and TF-IDF (both 0.0002), indicating the model effectively leveraged both hand-crafted and learned representations for optimal performance.

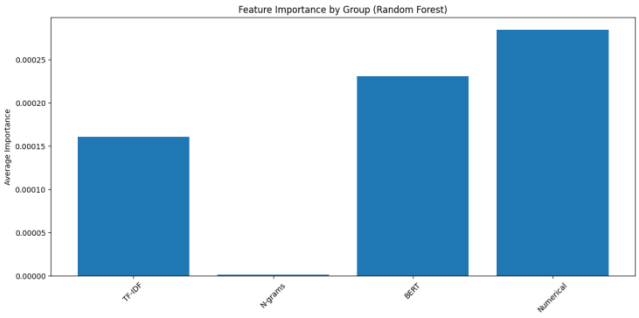


Figure 14.4 Feature Importance (Hybrid Random Forest)

4.3 Performance Comparison

The table below shows all the metrics we compared the models on:

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)
Naïve Bayes	0.8359	0.8520	0.8359	0.8322
Logistic Regression	0.9352	0.9364	0.9352	0.9344
Random Forest	0.9411	0.9423	0.9411	0.9413
XGBoost	0.9344	0.9354	0.9344	0.9340
Bi-LSTM	0.9294	0.9293	0.9294	0.9293
Hybrid BiLSTM-XGBoost	0.9550	0.9549	0.9550	0.9550
Hybrid BiLSTM-Random Forest	0.9517	0.9516	0.9517	0.9516

Table 2: Performance table for all models

A comprehensive comparison of all models reveals the clear superiority of the hybrid approaches, with both hybrid architectures consistently outperforming all individual models across every performance metric. The Hybrid BiLSTM-XGBoost achieved the highest overall performance with an accuracy of 0.9550, precision of 0.9549, recall of 0.9550, and F1-score of 0.9550, while the Hybrid BiLSTM-Random Forest achieved the highest AUC score of 0.9922. Among individual models, Random Forest demonstrated the strongest performance (accuracy: 0.9411), followed by Logistic Regression (0.9352) and XGBoost (0.9344), with all three significantly outperforming the Naïve Bayes baseline (0.8359). The performance hierarchy remained consistent across precision, recall, and F1-score metrics, confirming the robustness of the evaluation. The 2-3% performance improvement achieved by the hybrid models over the best individual model (Random Forest) represents a substantial advancement in detection capability, particularly significant for the challenging task of distinguishing between hate speech and offensive language.

The following are bar charts to visualize the comparison of the models based on different performance metrics (Accuracy, Precision, Recall, F1 Score, AUC):

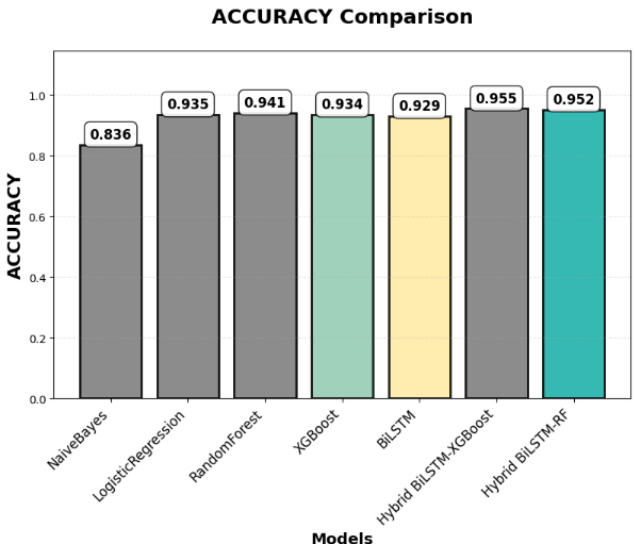


Figure 15.1 Accuracy Comparison across Models

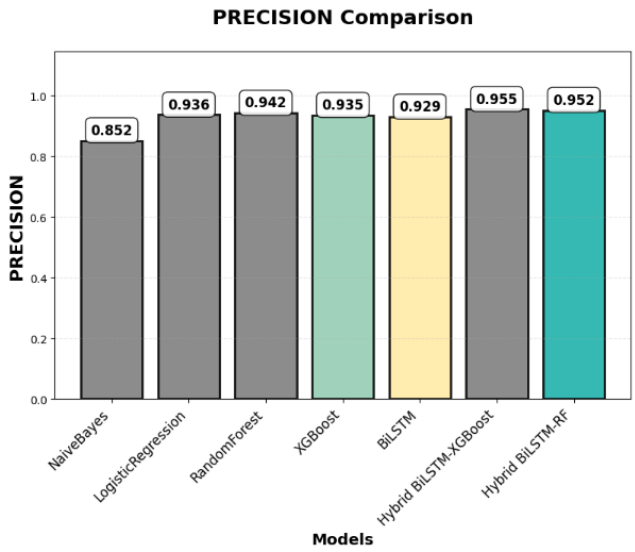


Figure 15.2 Precision Comparison across Models

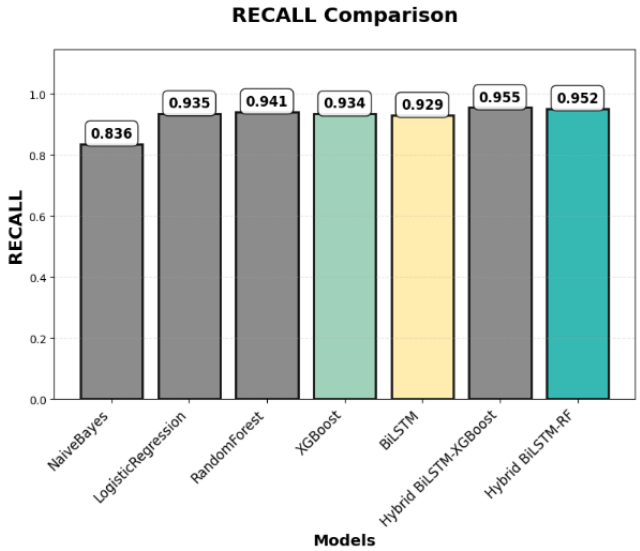


Figure 15.3 Recall Comparison across Models

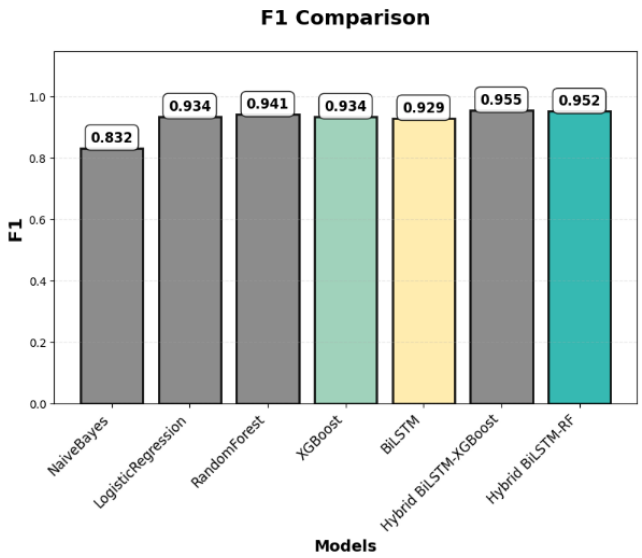


Figure 15.4 F1 Score Comparison across Models

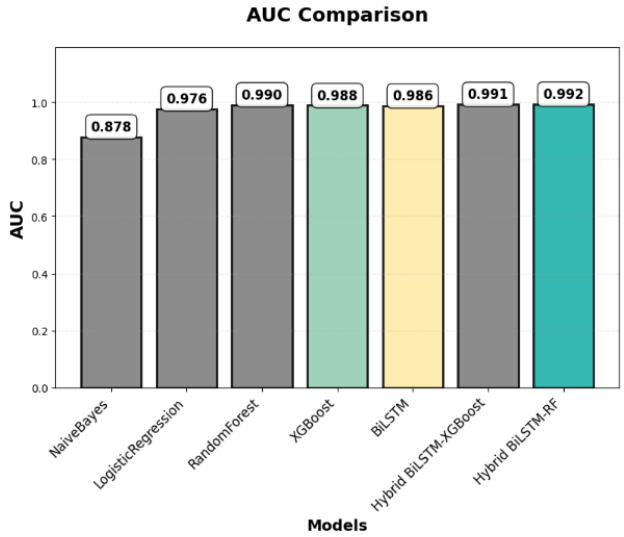


Figure 15.5 AUC Comparison across Models

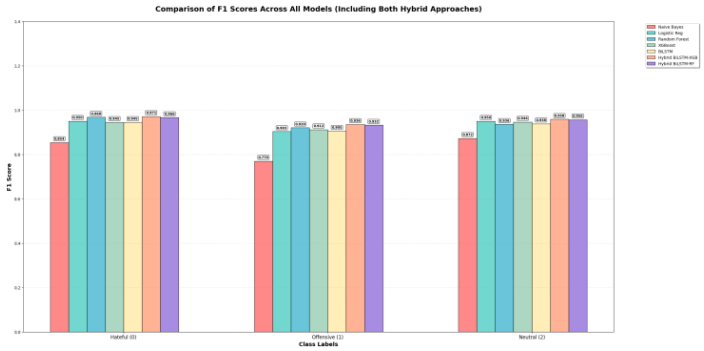


Figure 15.6 Comparison of F1 Scores per Class across All Models

4.4 Performance Comparison of Hybrid Models

A comparison of the two hybrid models reveals that although both significantly outperformed all individual models, the BiLSTM-XGBoost hybrid demonstrated a slight but consistent advantage over the BiLSTM - Random Forest approach. The BiLSTM - XGBoost achieved a marginally higher overall F1-score (0.9550 vs. 0.9516), representing a 0.34% performance difference. This superiority was consistent across all major metrics including accuracy, precision, and recall, though the Bi-LSTM - Random Forest hybrid achieved the highest AUC score (0.9922 vs. 0.9915). The performance advantage of the XGBoost-based hybrid suggests that its gradient boosting framework, which sequentially corrects errors from previous trees, may integrate more effectively with the deep features extracted by the Bi-LSTM than the bagging approach used by Random Forest. Despite this slight difference, both hybrid models demonstrated remarkable performance, with the minimal margin indicating that the core architectural advantage lies in the combination of deep learning feature extraction with ensemble classification, rather than the specific choice of tree-based algorithm.

The following is a bar chart that visualizes the comparison of the hybrid models based on F1-scores per class:

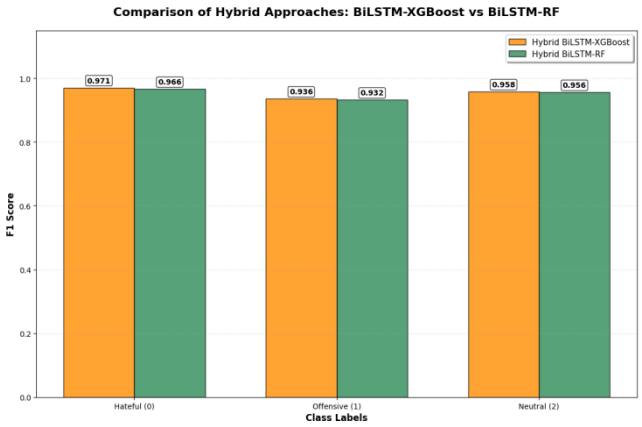


Figure 16 Comparison of F1 Score between Hybrid Models

4.5 Explainable AI

To move beyond the "black box" nature of complex hybrid models and build trust in their predictions, we employed SHAP (SHapley Additive exPlanations) values. SHAP is a unified framework based on cooperative game theory that assigns each feature an importance value for a particular prediction. This allows us to interpret the model's output globally (which features are most important on average) and locally (why a specific instance was classified a certain way).

4.5.1 Global Feature Importance

We conducted a global feature importance analysis for our two top-performing hybrid models: Hybrid BiLSTM - Random Forest (RF) and Hybrid BiLSTM - XGBoost. The results, highlighting the top 20 most impactful features, are presented in figures below:

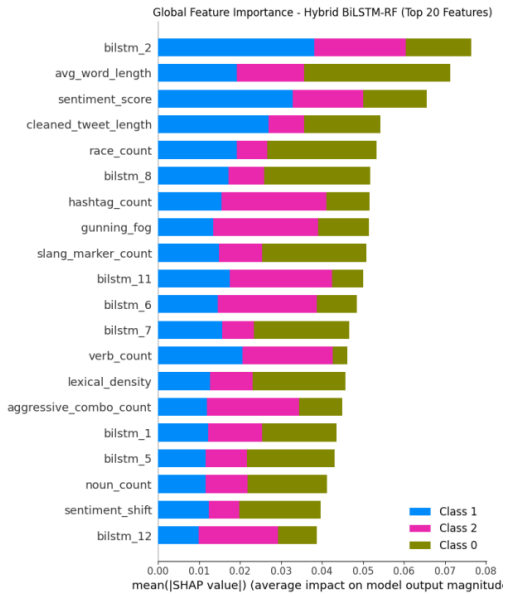


Figure 17.1 X-AI: Global Feature Importance of Bi-LSTM – Random Forest

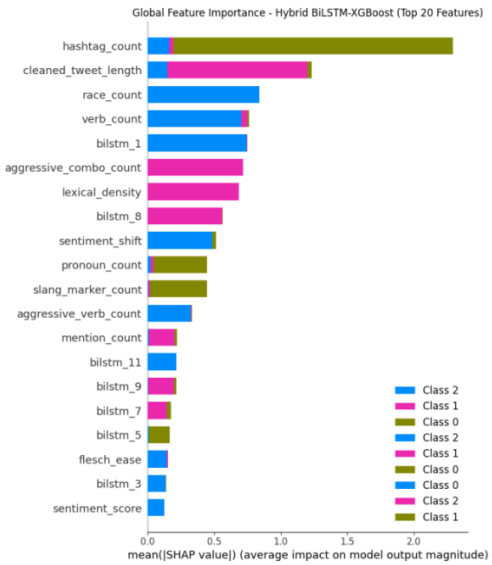


Figure 17.2 X-AI: Global Feature Importance of Bi-LSTM - XGBoost

The global interpretation reveals the features that, on average, have the largest impact on the model's output magnitude across all predictions.

For the BiLSTM-RF model, the most influential feature was biistm_2, a latent representation learned by the Bi-LSTM layer. This underscores the critical role of deep, sequential context in the classification task. This was closely followed by fundamental text statistics like avg_word_length and sentiment_score, as well as structural features like cleaned_tweet_length and race_count. The high ranking of several other Bi-LSTM components (biistm_8, biistm_11, etc.) confirms that the model heavily relies on the nuanced semantic features extracted by the deep learning component, which are then effectively leveraged by the Random Forest classifier. Readability metrics (gunning_fog) and stylistic markers (slang_marker_count) also featured prominently.

On the other hand, BiLSTM - XGBoost architecture showed a slight difference of feature prioritization. In this scenario, hashtag_count emerged as the most important feature, suggesting that the use of hashtags is a very significant marker for the XGBoost classifier. Structural and lexical features like cleaned_tweet_length, race_count, verb_count, and lexical_density were also among the top contributors. While BiLSTM features (biistm_1, biistm_8) remained highly important, their relative ranking was lower compared to the RF-based model, indicating that XGBoost places a stronger emphasis on the hand-crafted linguistic and metadata features. The presence of aggressive_combo_count and aggressive_verb_count highlights the model's focus on specific aggressive language cues.

4.5.2 Model Consensus and Insights

Despite differences in ranking, both models consistently identified a core set of feature types as highly important:

- Deep Semantic Features: Bi-LSTM latent representations were among the top features for both models, proving their value in capturing context beyond what traditional NLP features can.
- Structural Metadata: Features like hashtag_count, cleaned_tweet_length, and mention_count are strong indicators of user engagement style, which is highly relevant for the task.
- Linguistic Content: The importance of sentiment_score, sentiment_shift, aggressive_combo_count, and various POS counts (verb_count, noun_count) confirms that the models are indeed leveraging linguistic signals related to emotion and aggression, aligning well with the objectives of this research.
- Lexical Complexity: Features such as lexical_density and gunning_fog suggest that the complexity and sophistication of the language used also play a role in the classification.

4.5.3 Class-Specific Impact

The color coding in the SHAP plots (Class 0, Class 1, Class 2) illustrates the direction of a feature's impact for each class. For instance, a high sentiment_score (positive sentiment) might push the prediction towards a non-aggressive class (e.g., Class 0), resulting in a high SHAP value for that class, while simultaneously pulling it away from aggressive classes (Class 1 or 2).

4.6 Insights

The research shows that the Hybrid Bi-LSTM – XGBoost is the best performing model out of all the models evaluated, with an F1-score of 99.50%, slightly edging over the Hybrid Bi-LSTM – Random Forest model by 0.34%. Both the hybrid architectures, with their performance metrics gave credence to the initial intuitions and hypotheses of the research that hybrid, deep ensemble learning techniques can provide significant promise of complex sentiment classification tasks like hate speech detection in terms of better precision for 'Hate Speech' (97% on Hybrid Bi-LSTM – XGBoost) to distinguish it from 'Offensive Language'.

5. CONCLUSION

This research was an attempt to address and come up with a refined solution to the issue of distinguishing between offensive language and actual hate speech on social media. The proposed hybrid ensemble architectures that combined the synergy of Bi-Directional LSTM's capability to deal with sequential texts and the leveraging capabilities of ensemble models like XGBoost and Random Forest to combine the contextual learning of Bi-LSTM and the comprehensive suite of features of BERTweet embeddings, TF-IDF, n-grams and hand-crafted linguistic features like slang-markers, NER, lexical density, sentiment shifts, readability scores etc. that attempt to dissect especially complicated parts of casual language like urban slangs, coded languages and veiled threats.

The research, evaluated on the benchmark Crowdflower[1] dataset, shows that our hybrid models demonstrated state-of-the-art performance, especially the Hybrid Bi-LSTM – XGBoost architecture which proved to be the most effective, achieving a superior F1-score (95.50%); and significantly enhances the precision, recall, F1-scores for 'Hateful' speech (97% for each). This represents a substantial improvement over all the individual models, confirming the initially proposed hypothesis that deep learning and ensemble techniques would provide a better solution for the ternary classification task. Furthermore, do deal with the 'Black-Box' nature of such hybrid architectures, SHAP-based Explainable AI (XAI) analysis gave a crucial interpretability factor to the hybrid models and verified that the predictions are driven by a balanced contribution of contextual features from deep-learning and established linguistic characteristics, such as sentiment scores, aggression markers. Slang markers etc. as well as metadata features like hashtag and mention counts – rather than imbalanced correlations of one type of feature. To conclude, this work provides a solid framework that is accurate, robust and interpretable for automated hate speech detection. The performance that has been demonstrated are essential for the ethical deployment of such systems for content moderation in real-world online platforms in avoiding harmful conduct while also minimizing unjust censorship.

6. FUTURE WORK

- Integration with datasets from other platforms (such as chat-logs, comment sections from various public servers from Facebook, Instagram, Discord, Reddit, Quora etc.), larger as well as multimodal datasets combining image and sound data, as social media platforms nowadays are endlessly expandable with audio-visual data dominating the scenario. Furthermore, including different forms of hate speech such as 'Harassment', 'Bullying', 'Racism', 'Sexism', 'Xenophobia', 'Bigotry', 'Fascism' etc. as well as being able to classify non-consensual advances such as different threats of 'Predatory', 'Pedophilic', 'Sexual Assault' etc. for better identification of the type of offense that is being threatened in the speech.
- Exploration of other deep learning models like GRU, RNN, CNN and transformers like BERT and different versions of it like RoBERT, DistilBIRT and seeing how they fair if combined with similar ensemble techniques.
- Improved feature extraction for better distinguishability between 'Hateful' and 'Offensive', such as implementing LDA, other contextual embeddings (Word2Vec, GloVe etc.), other metadata features like tweet timing, reply velocity etc. as well as additional hyperparameter tuning to make sure the model gets the most from deep learning.

7. REFERENCES

[1] Davidson, T., Warmsley, D., Macy, M. and Weber, I. 2017. Automated hate speech detection and the problem of offensive language. *arXiv (Cornell University)*. DOI: https://doi.org/10.48550/arxiv.1703.04009.

[2] Watanabe, H., Bouazizi, M. and Ohtsuki, T. 2018. Hate Speech on Twitter: A pragmatic approach to collect hateful and offensive expressions and perform hate speech detection. *IEEE Access*. 6, 13825–13835. DOI: https://doi.org/10.1109/access.2018.2806394.

[3] Mungoli, N. 2023. Adaptive Ensemble Learning: Boosting Model Performance through Intelligent Feature Fusion in Deep Neural Networks. *arXiv (Cornell University)*. DOI: https://doi.org/10.48550/arxiv.2304.02653.

[4] Singh, D., Bhattacharjee, R., and Chakraborty, A. 2025. Rethinking Hate Speech Detection on Social Media: Can

LLMs Replace Traditional Models? *arXiv*. DOI: https://doi.org/10.48550/arXiv.2506.12744

[5] Zampieri, N., Illina, I., and Fohr, D. 2021. Improving Automatic Hate Speech Detection with Multiword Expression Features. *arXiv*. DOI: https://doi.org/10.48550/arXiv.2106.00237

[6] Su, K., Wu, J., Gu, D., Yang, S., Deng, S., and Khakimova, A. K. 2021. An Adaptive Deep Ensemble Learning Method for Dynamic Evolving Diagnostic Task Scenarios. *Diagnostics*, 11, 12, 2288. DOI: https://doi.org/10.3390/diagnostics11122288

[7] Perera, S., Meedin, N., Caldera, M., Perera, I., and Ahangama, S. 2023. A comparative study of the characteristics of hate speech propagators and their behaviours over Twitter social media platform. *Heliyon*, 9, 8, e19097. DOI: https://doi.org/10.1016/j.heliyon.2023.e19097

[8] Davidson, T., Bhattacharya, D., and Weber, I. 2019. Racial bias in hate speech and abusive language detection datasets. *Proceedings of the Third Workshop on Abusive Language Online*, 25–35. DOI: https://doi.org/10.48550/arXiv.1905.12516

[9] Bird, S., Klein, E., & Loper, E. 2009. *Natural language processing with Python.* O'Reilly Media.

[10] Hutto, C. and Gilbert, E. 2014. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*. 8, 1, 216–225. DOI: https://doi.org/10.1609/icwsm.v8i1.14550.

[11] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Müller, A., Nothman, J., Louppe, G., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, É. 2011. Scikit-learn: Machine Learning in Python. *arXiv*. DOI: https://doi.org/10.48550/arXiv.1201.0490

[12] Agarwal, S. and Chowdary, C. R. 2021. Combating hate speech using an adaptive ensemble learning model with a case study on COVID-19. *Expert Systems with Applications*, 185, 115632. DOI: https://doi.org/10.1016/j.eswa.2021.115632

[13] Nguyen, D. Q., Vu, T., and Nguyen, A. T. 2020. BERTweet: A pre-trained language model for English Tweets. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 9–14. DOI: https://doi.org/10.18653/v1/2020.emnlp-demos.2

[14] Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321–357. DOI: https://doi.org/10.1613/jair.953

[15] Breiman, L. 2001. Random Forests. *Machine Learning*, *45*, 1, 5–32. DOI: https://doi.org/10.1023/a:1010933404324

[16] Zhao, C., You, J., Wen, X., and Li, X. 2020. Deep Bi-LSTM networks for sequential recommendation. *Entropy*, 22, 8, 870. DOI: https://doi.org/10.3390/e22080870

[17] Fan, Y., Tang, Q., Guo, Y., & Wei, Y. 2024. BILSTM-MLAM: a Multi-Scale Time Series prediction model for sensor data based on Bi-LSTM and local attention mechanisms. *Sensors*, 24, 12, 3962. DOI: https://doi.org/10.3390/s24123962

[18] Koishigarina, D., Uselis, A., and Oh, S. J. 2025. CLIP Behaves like a Bag-of-Words Model Cross-modally but not Uni-modally. *arXiv*. DOI: https://doi.org/10.48550/arXiv.2502.03566